

Quantifying Arroyo’s Asthenocoria Sign: A Low-Cost Embedded Pupillography System for Naturopathic Clinical Screening

Walker Kirkpatrick, ND

September 2026

Abstract

Arroyo’s asthenocoria sign, first described in 1924, is the failure of the pupil to sustain constriction under continuous light exposure—a phenomenon historically associated with adrenal insufficiency in naturopathic and functional medicine. Despite nearly a century of clinical use, the test has remained a subjective bedside observation dependent on examiner skill and patient cooperation. We present **arroyo-pi**, an open-source embedded pupillography system that quantifies the Arroyo sign using a 940 nm near-infrared LED, a NoIR camera module (Arducam IMX462 for Raspberry Pi or IMX219-NoIR for Jetson Orin Nano), and a Python analysis core built on OpenCV dark-pupil detection. The system records pupil diameter via the pupil-to-iris ratio (PIR) method, calibrated against a horizontal visible iris diameter of 11.7 mm, and applies a deterministic analysis pipeline: baseline median estimation, maximum constriction search, escape detection at 15% of constriction amplitude sustained for 2 s, oscillation counting with running-median detrending, and four-band grading (healthy ≥ 20 s, mild 10–20 s, fatigue 5–10 s, exhaustion < 5 s). A suite of 25 unit tests with synthetic pupillograms validates all grading bands, edge cases, and symmetry comparisons. The system is the hardware successor to an Apple Vision Pro prototype, replacing proprietary camera restrictions with a sub-\$100 embedded platform. We frame the tool honestly: it measures pupillary behavior, not adrenal function, and the grading thresholds are functional-medicine screening bands, not clinically validated diagnostic criteria. The system enables objective trend tracking and removes observer bias from a test that has lacked instrumentation for a century.

1 Introduction

The pupillary light reflex is among the most accessible windows into autonomic nervous system function. A clinician can observe it with nothing more than a penlight and a darkened room. Yet this accessibility has paradoxically constrained its clinical utility: without instrumentation, the observation remains qualitative, subject to inter-examiner variability, and impossible to track longitudinally with any rigor.

Arroyo’s asthenocoria sign—the failure of the pupil to hold constriction under sustained light—exemplifies this gap. Described in 1924 by Carlos F. Arroyo as a correlate of adrenal insufficiency, the sign has persisted in naturopathic and functional medicine practice for a century without ever transitioning from subjective observation to objective measurement. A practitioner shines a light across the patient’s eye, watches the pupil constrict, and then waits to see whether constriction is maintained or the pupil redilates despite continued illumination. The result is a judgment call, graded by a stopwatch at best and by impression at worst.

This paper presents **arroyo-pi**, a low-cost embedded pupillography system that replaces the subjective exam with instrumented measurement. The system uses near-infrared illumination invisible to the human eye, a consumer-grade NoIR camera, and an OpenCV-based tracking pipeline to record pupil diameter continuously throughout the test. A Python analysis core then applies deterministic algorithms to detect pupillary escape, quantify hold time, and assign a grade. The hardware platform is a Raspberry Pi 4/5 or NVIDIA Jetson Orin Nano, making the system accessible to clinics that cannot justify research-grade pupillometry equipment.

We position this system as a screening and trend-tracking instrument, not a diagnostic device. The physiological relationship between sustained pupillary behavior and adrenal function is not established by conventional endocrinology, and the grading thresholds used in functional medicine practice have not been clinically validated. What the system provides is objectivity, repeatability, and freedom from observer bias—enabling practitioners to track changes over time with quantitative confidence.

2 Historical Background

Carlos F. Arroyo (1892–1928) was a physician whose 1924 publication in the *Medical Journal and Record* (January 2, 1924) described a pupillary sign he observed in patients with adrenal insufficiency. Arroyo termed this sign *asthenocoria*—from the Greek *astheneia* (weakness) and *kore* (pupil)—characterizing it as a sluggishness or weakness of the pupillary light reflex under sustained illumination [3,4].

The Farlex Medical Dictionary defines asthenocoria as “sluggishness of the pupillary light reflex,” attributing the eponym *Arroyo sign* to the same phenomenon [3]. The IQB Spanish Medical Dictionary similarly records the *Signo de ARROYO* as a sluggish pupillary reaction to light [4].

Arroyo’s original description, as preserved in the Thyroid Patient Advocacy UK source [1], reads:

When exploring the pupil area reflex, I found that in the iris of those cases (adrenal insufficiency), although reacting readily to light, the contraction (of the iris) was flabby, lazy, in a word asthenia. By making the patient look at the light we see that immediately after the initial mitosis the pupil starts to dilate slowly as if it does not want to, seems to try to contract again but the dilation gains the upper hand and, after a fight between miosis and mydriasis lasting for about 40 seconds, the pupil remains dilated in spite of the persistence of the exciting agent (the light). This sign is consistent and present in all cases of hypoadrenia in all of its clinical forms. In the normal individual, it does not appear as I have investigated. All patients presenting this sign, which I should like to call asthenocaria, have been benefited by suprarenal medication.

The protocol Arroyo described is straightforward: darken the room, shine a light across one eye (not directly into it), and observe whether the initial constriction is maintained or the pupil redilates despite continued illumination. The sign is positive when redilation or oscillation occurs under sustained light. Arroyo reported that the sign was absent in healthy individuals and present in all clinical forms of hypoadrenia he investigated [1].

3 Naturopathic and Functional Medicine Context

Arroyo’s test survived in naturopathic and functional medicine literature long after it faded from conventional practice. The test is commonly known in these traditions as the *iris contraction test* or the *pupil response test*, and it appears in practitioner guides and patient-facing resources as a self-administered screening tool for adrenal dysfunction [2,23,24,25].

James L. Wilson coined the term “adrenal fatigue” in 1998 to describe a putative syndrome of subclinical adrenal hypofunction characterized by fatigue, salt craving, and impaired stress tolerance [22]. Within this framework, Arroyo’s iris contraction test became a standard screening instrument. Wilson’s framework proposed that under chronic stress, the adrenal glands progress through stages of alarm, resistance, and exhaustion, and that pupillary instability reflects this progressive decompensation [22,30].

The grading thresholds most commonly cited in functional medicine literature are [2,23]:

- **Healthy:** constriction held for 20 seconds or more
- **Mild disruption:** constriction held for 10–20 seconds
- **Fatigue:** constriction held for 5–10 seconds
- **Exhaustion:** constriction fails within 5 seconds

Some protocol variants specify the light source at 8 inches and 45 degrees from the eye [5], while others recommend a penlight held approximately 20 cm from the eye at a similar angle. The common elements are a dim room, a sustained lateral light source, and timed observation of pupil behavior.

The test also appears in syntonics optometry, a branch of optometric practice that uses light therapy. The *Alpha Omega Pupil* concept in syntonics optometry relates pupillary response to autonomic balance and visual field constriction, drawing on similar observations of sustained constriction capacity [6].

We emphasize that these thresholds are **not** clinically validated. They originate from clinical observation and practitioner tradition, not from controlled studies with defined populations. The test functions as a screening and trend instrument: a practitioner uses it to track changes within an individual over time, not to diagnose a specific pathology. The system described in this paper preserves these thresholds as configurable parameters precisely because they represent the conventional practice the system is designed to instrument.

4 Physiological Basis

4.1 The Pupillary Light Reflex

The pupillary light reflex (PLR) is the constriction of the pupil in response to retinal illumination, mediated by a subcortical reflex arc. Light striking the retina activates retinal ganglion cells whose axons travel via the optic nerve to the pretectal nuclei in the midbrain. From there, signals reach the Edinger–Westphal nuclei, whose parasympathetic fibers travel via the oculomotor nerve to the sphincter pupillae muscle, producing constriction [7,8,27].

Pupil diameter is governed by the balance between parasympathetic constriction (sphincter pupillae) and sympathetic dilation (dilator pupillae). The parasympathetic pathway acts rapidly: light onset triggers constriction within approximately 200–500 ms. The sympathetic pathway acts more slowly, modulating baseline pupil size and contributing to redilation after light offset [7].

4.2 Pupillary Escape as a Normal Phenomenon

Pupillary escape—the partial redilation of the pupil despite continued light stimulation—is a **normal** physiological phenomenon, not inherently pathological. Loewenfeld and her collaborators established that under sustained illumination, the pupil initially constricts and then partially redilates

as the retinal response adapts [26]. This escape reflects the adaptation of the retinal photoreceptors and the dynamics of the neural integpathway, not a failure of any specific organ system.

The degree of pupillary escape depends on stimulus intensity, adaptation state, and individual variability. Under bright light, constriction is well sustained; under dimmer or moderate light, escape is more pronounced [26,27]. This is a critical consideration for Arroyo’s test: the protocol uses a penlight at an angle, producing a moderate and somewhat variable stimulus, which makes the amount of escape highly sensitive to stimulus conditions.

4.3 Melanopsin and Intrinsically Photosensitive Retinal Ganglion Cells

A subset of retinal ganglion cells—intrinsically photosensitive retinal ganglion cells (ipRGCs)—contain melanopsin, a photopigment with peak sensitivity near 480 nm. These cells contribute to sustained pupillary constriction under continuous light, providing a tonic constriction signal that complements the phasic response of rod and cone pathways [26,27].

The melanopsin contribution helps explain why the pupil can maintain partial constriction under prolonged illumination. However, the sustained-light observation used in Arroyo’s test differs fundamentally from flash pupillometry: flash techniques measure the transient response to brief stimuli, while Arroyo’s protocol observes the sustained response over tens of seconds, where adaptation, escape, and oscillation dynamics dominate.

4.4 The Four Phases of the Pupillary Light Reflex

The PLR under sustained light can be decomposed into four phases [8,27]:

1. **Latency:** The delay between light onset and the beginning of constriction, typically 200–500 ms. The `arroyo-pi` system detects latency as the first frame where pupil diameter drops 5% below baseline, sustained for 3 consecutive samples.
2. **Constriction:** The active narrowing phase, reaching maximum constriction within 1–2 s of light onset. The system searches for minimum pupil diameter within a 5 s post-stimulus window.
3. **Escape:** Partial redilation despite continued illumination, reflecting retinal adaptation and the balance of autonomic inputs. The system flags escape when the pupil redilates by 15% or more of the constriction amplitude, sustained for at least 2 s.
4. **Recovery:** Return toward baseline after light offset, driven by sympathetic dilation and cessation of parasympathetic tone.

5 Clinical Limitations and Honest Framing

We are deliberate about what this system does and does not measure.

“Adrenal fatigue” is not a recognized diagnosis in conventional endocrinology. The Endocrine Society and other major medical bodies do not recognize it as a valid clinical entity, and the symptoms attributed to it overlap substantially with other conditions [29,31]. WebMD notes that “adrenal fatigue” as a concept lacks the scientific evidence required for clinical acceptance [29]. Medical News Today similarly reports that the term is not supported by endocrinological research [31].

The gold standard for diagnosing adrenal insufficiency is the ACTH (cosyntropin) stimulation test, which measures cortisol response to exogenous adrenocorticotrophic hormone [10]. This is

a blood test performed in a clinical laboratory, not a pupillary observation. Primary adrenal insufficiency (Addison’s disease) is rare and requires definitive endocrinological workup [10].

Asthenocoria itself is nonspecific. The Ulyclinic clinical reference notes that Arroyo’s sign can appear in various conditions and is not specific to adrenal dysfunction [9]. Pupillary escape is influenced by fatigue, lighting conditions, medications, autonomic tone, age, and individual variability. A positive Arroyo sign does not indicate adrenal pathology, and a negative sign does not rule it out.

The arroyo-pi system measures pupillary behavior—not adrenal function. It quantifies how long the pupil sustains constriction under continuous light, how much it redilates, and whether it oscillates. These are pupillary measurements. Any inference about adrenal status is a clinical judgment made by the practitioner, not a measurement made by the system.

The grading thresholds (20 s, 10 s, 5 s) are functional-medicine screening bands, not validated diagnostic criteria. They are useful for tracking trends within an individual over time, which is how naturopathic practitioners typically use the test [24,25]. We preserve these thresholds in the system as configurable parameters, and we explicitly label every report with a reminder that the system is a screening instrument.

6 System Architecture

6.1 Hardware

The **arroyo-pi** system consists of three hardware components (Figure 3):

- **IR illumination:** A 940 nm near-infrared LED provides illumination for the camera. The 940 nm wavelength is invisible to the human eye and does not trigger the pupillary light reflex, allowing the camera to observe the pupil without itself altering pupil size [18,19]. This is a critical design choice: visible illumination would confound the test by acting as an uncontrolled stimulus. The 940 nm band is widely used in eye-tracking systems for this reason [18,19].
- **Camera:** A NoIR (no infrared filter) camera module captures the eye under IR illumination. Two options are supported:
 - **Raspberry Pi:** Arducam IMX462 Pivariety camera, a low-light-optimized sensor with a Starvis sensor suitable for IR imaging [17].
 - **Jetson Orin Nano:** IMX219-NoIR, the standard Raspberry Pi camera module V2 with the IR filter removed, connected via the Jetson’s MIPI-CSI interface.

Both cameras operate at 30 fps at 640×480 resolution, which is sufficient for the temporal dynamics of the Arroyo test (pupillary changes occur over seconds, not milliseconds).

- **Compute:** Raspberry Pi 4 or 5, or NVIDIA Jetson Orin Nano. The Jetson provides GPU acceleration if needed for future deep-learning-based tracking, but the current OpenCV pipeline runs comfortably on CPU at 30 fps.

The total hardware cost is under \$100, compared to thousands of dollars for research-grade pupillometry equipment [15,21].

6.2 Software

The software stack is implemented in Python with OpenCV for computer vision:

- **Pupil tracking** (`tracker.py`): Dark-pupil detection under 940 nm IR. The pipeline uses Haar cascade classifiers for face and eye region detection, Hough circle fitting for the iris boundary, and adaptive thresholding with contour analysis and ellipse fitting for the pupil. The pupil is identified as the darkest circular region within the eye ROI, scored by circularity, darkness, and size [16].
- **PIR calibration**: The pupil-to-iris ratio (PIR) method converts pixel measurements to millimeters. The iris diameter serves as an internal calibration ruler, with the horizontal visible iris diameter (HVID) assumed to be 11.7 mm (adult average). PIR is computed as pupil diameter divided by iris diameter, both in pixels, and multiplied by 11.7 mm to obtain absolute pupil diameter. This approach avoids the need for external calibration and is robust to changes in camera distance [27].
- **Analysis core** (`analysis.py`): A Python-native implementation of the Arroyo analysis algorithm, ported from the `arroyo-visionos` reference core. The algorithm detects pupillary escape, counts oscillations, and assigns a grade based on hold time. Section 7 describes the pipeline in detail.
- **CLI** (`main.py`): A command-line interface with three modes: `live` (camera exam), `demo` (synthetic data for testing without hardware), and `calibrate` (camera preview with tracking overlay for positioning).

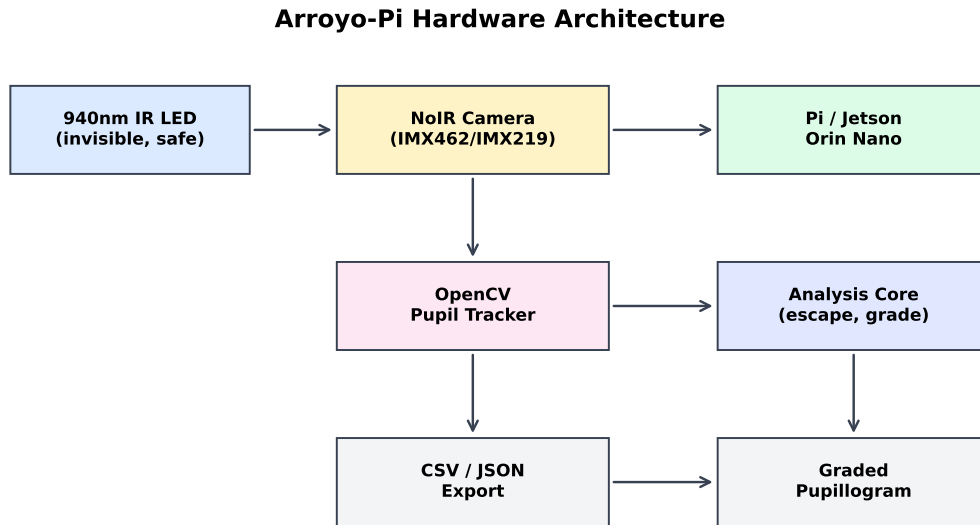


Figure 1: Hardware architecture block diagram. The 940 nm IR LED provides invisible illumination; the NoIR camera captures the eye; the Raspberry Pi or Jetson processes frames in real time with OpenCV dark-pupil detection.

7 Algorithm Details

The analysis pipeline processes a time series of pupil measurements and produces a graded result. Each step is implemented in `analysis.py` and is deterministic—no machine learning or stochastic components are involved. The algorithm operates on PIR (pupil-to-iris ratio) values rather than raw pixel counts, providing scale invariance.

7.1 PIR Interpolation of Invalid Frames

The tracker marks frames as invalid when no pupil is detected (e.g., during blinks or tracking loss). Invalid frames are linearly interpolated between the nearest valid neighbors to produce a continuous PIR time series. If fewer than 50% of frames are valid, the recording is rejected as insufficient data.

7.2 Baseline Estimation

The baseline PIR is computed as the median of all valid samples in the 2-second window preceding stimulus onset (t_0). If no pre-stimulus data is available (e.g., the recording starts at or after stimulus onset), the first 2 s of the recording are used and a quality warning is emitted. The median is chosen over the mean for robustness against outliers and blink artifacts.

7.3 Maximum Constriction Search

The minimum PIR (maximum constriction) is found by searching the 5-second window following stimulus onset ($t_0 \leq t \leq t_0 + 5$ s). The constriction amplitude is the difference between baseline PIR and minimum PIR. If the amplitude is less than 5% of baseline, the recording is graded as “no reflex”—no reliable constriction was detected.

7.4 Escape Detection

Escape is defined as redilation reaching 15% of the constriction amplitude, sustained for at least 2 s. The algorithm scans the post-constriction time series and, for each point where the relative redilation exceeds 15%, checks whether the redilation is maintained above 15% for the subsequent 2 s. The first such sustained crossing defines the escape time. The hold time is the interval from maximum constriction to escape onset. If no escape is detected, the pupil is graded as “held” (constriction maintained for the full recording).

7.5 Oscillation Counting

Oscillations—rhythmic fluctuations in pupil diameter after constriction—are counted using a de-trending and hysteresis approach:

1. The post-constriction PIR series is detrended using a running median with a 3-second window, removing slow drift while preserving oscillations.
2. A hysteresis band of $\pm 10\%$ of constriction amplitude is applied to the residual. The algorithm tracks direction state and counts zero-crossings of the hysteresis band, with each complete cycle (up-crossing followed by down-crossing) counted as one oscillation.
3. The oscillation frequency is the cycle count divided by the post-constriction duration.

Oscillations at 0.3 Hz or above, combined with rapid escape, characterize the exhaustion band.

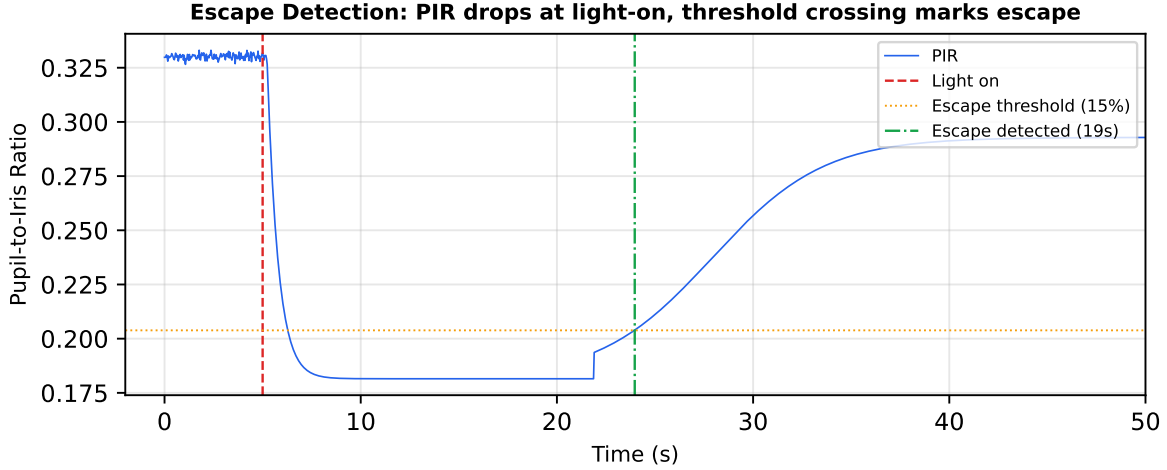


Figure 2: PIR time series showing the escape detection algorithm. The baseline median, maximum constriction, and 15% escape threshold are annotated. The pupil redilates past the threshold and sustains for > 2 s, triggering an escape event.

7.6 Grading Logic

The grade is assigned based on hold time (time from maximum constriction to escape) and oscillation behavior:

- **Held / Healthy (≥ 20 s):** No escape detected, or constriction maintained for the full recording. Negative Arroyo sign.
- **Mild (10–20 s):** Escape after 10–20 s of sustained constriction.
- **Fatigue (5–10 s):** Escape after 5–10 s of sustained constriction.
- **Exhaustion (< 5 s):** Constriction fails within 5 s, or immediate pulsation with oscillations ≥ 0.3 Hz.

For two-eye sessions, the overall grade is the worse of the two eyes, and symmetry metrics (hold time difference, constriction amplitude difference) are reported.

8 Validation

The analysis core is validated by a suite of 25 unit tests (`tests/test_analysis.py`) using synthetic pupillograms with known ground truth. The synthetic generator (`synth_eye`) produces deterministic pupillograms parameterized by escape hold time, oscillation frequency, constriction fraction, and noise level, allowing each grading band and edge case to be tested independently.

The test suite covers:

- **Synthetic generator verification** (3 tests): Valid generation of healthy and fatigue patterns; blink frame invalidation.
- **Healthy pupil** (4 tests): No escape detected; constriction percentage in expected range (30–55%); valid baseline diameter; holding index > 0.80 .

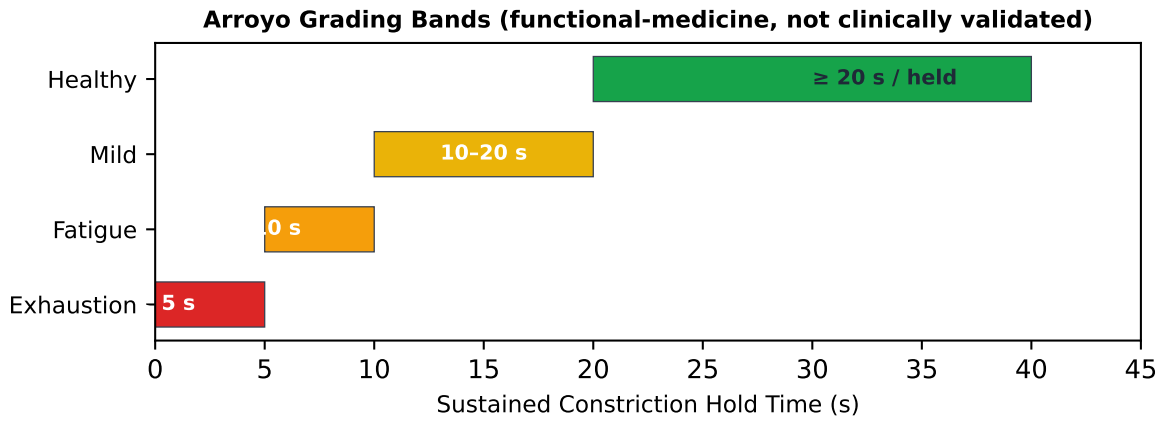


Figure 3: Grading threshold bands. Hold time is mapped to four functional bands: healthy (≥ 20 s), mild (10–20 s), fatigue (5–10 s), and exhaustion (< 5 s). These are screening bands, not validated diagnostic thresholds.

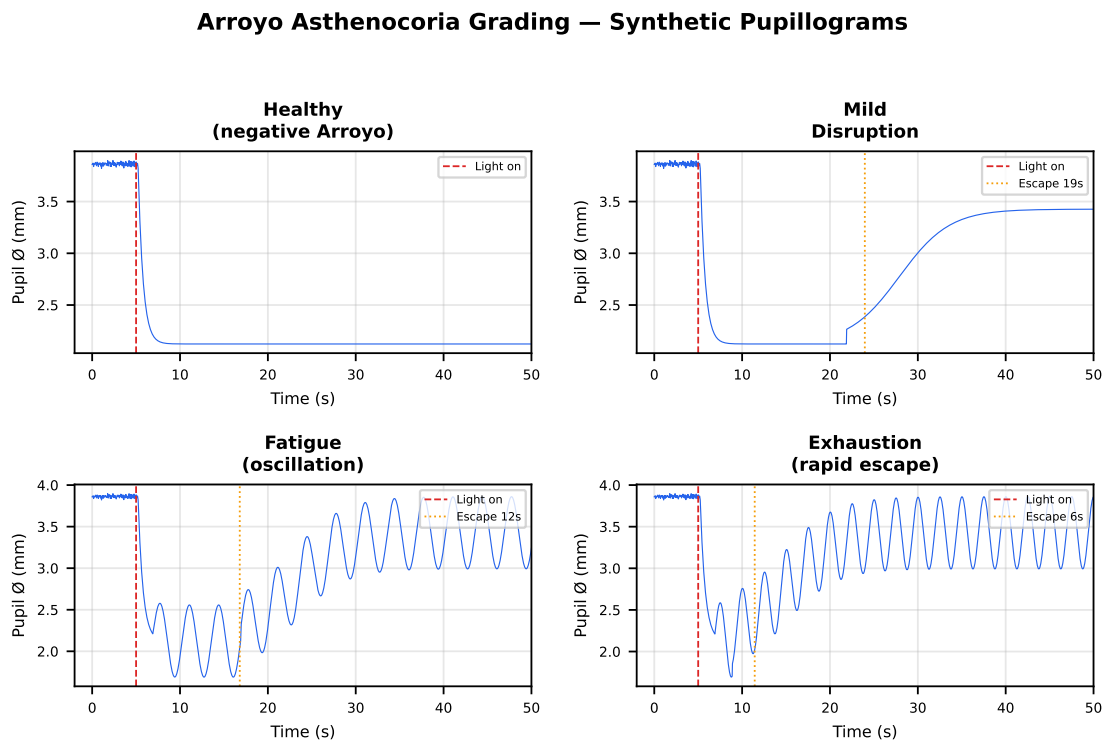


Figure 4: Four-grade pupillogram comparison. Synthetic pupillograms illustrate each grade: healthy (sustained constriction), mild (late escape), fatigue (early escape with oscillation), and exhaustion (rapid failure).

- **Mild escape** (3 tests): Escape detected; grade assigned as “mild”; escape time within the 12–22 s band.
- **Fatigue with oscillation** (3 tests): Escape detected; grade “fatigue”; oscillation count > 5 and frequency > 0.1 Hz.
- **Exhaustion** (3 tests): Escape detected; grade “exhaustion”; escape time < 8 s.
- **No reflex** (1 test): Flat pupil (0.1% constriction) graded as “noReflex”.
- **Insufficient data** (3 tests): Empty sample list; too few samples (< 5); majority-invalid frames ($< 50\%$ valid) all graded as “insufficientData”.
- **Session symmetry** (3 tests): Both eyes healthy $\rightarrow$ overall healthy; disagreeing eyes $\rightarrow$ overall takes worse grade; both fatigue $\rightarrow$ overall fatigue.
- **Quality warnings** (2 tests): Low sample rate (< 15 fps) triggers warning; short recording (< 30 s) triggers warning.

All 25 tests pass. The tests are run with Python’s `unittest` framework and require no external hardware, making them suitable for continuous integration. The synthetic generator’s deterministic seeding (random seed parameter) ensures reproducibility across test runs.

9 Discussion

9.1 What This System Enables

For naturopathic and functional medicine practitioners, the `arroyo-pi` system transforms Arroyo’s test from a subjective observation into a quantitative measurement. The key benefits are:

- **Objective trend tracking:** A practitioner can record a pupillogram at each visit and compare hold times, constriction amplitudes, and oscillation patterns over months or years. Changes are quantified in seconds and percentages, not described as “better” or “worse.”
- **Repeatability:** The 940 nm IR illumination is invisible to the patient, eliminating the confound of the measurement light itself affecting the pupil. The PIR calibration method is robust to small changes in camera positioning between sessions.
- **Removal of observer bias:** The algorithm applies identical criteria to every recording. Two practitioners analyzing the same pupillogram will get the same grade, because the grading is deterministic.
- **Documentation:** Each exam produces a CSV of the raw pupillogram and a JSON report with all metrics, creating a longitudinal record that can be reviewed, shared, or audited.

9.2 Comparison to the Apple Vision Pro Approach

The `arroyo-pi` system is the hardware successor to `arroyo-visionos`, an Apple Vision Pro application that implemented the same analysis algorithm using the headset’s built-in eye-tracking cameras. The Vision Pro approach had significant advantages: high-quality eye tracking, controlled IR illumination, and a self-contained headset form factor. However, Apple’s enterprise camera restrictions and the \$3,49 device cost made it impractical for clinical deployment [12].

The Raspberry Pi / Jetson approach sacrifices some tracking precision but gains accessibility. A practitioner can assemble the system for under \$100 using off-the-shelf components. The analysis algorithm is identical between the two platforms—only the data acquisition layer differs.

9.3 Comparison to Research-Grade Pupillometers

OpenIris is an open-source eye-tracking framework that achieves 500 Hz sampling with FLIR machine-vision cameras, providing millisecond-resolution pupillary dynamics [15,21]. The MEYE-Lens system is an open-source pupillometry eyewear platform for mobile measurement [20]. Smartphone-based pupillometry approaches have also been validated against clinical instruments [11,12].

The **arroyo-pi** system operates at 30 fps with consumer cameras, two orders of magnitude slower than OpenIris. However, the Arroyo test operates on a timescale of seconds, not milliseconds. The critical measurements are hold time (seconds), escape threshold crossing (sustained over 2 s), and oscillation frequency (~ 0.3 Hz). A 30 fps sampling rate provides 33 ms temporal resolution, which is more than adequate for these dynamics. The trade-off is clear: the system exchanges high-speed precision for low-cost accessibility, and for the Arroyo test specifically, this is a reasonable exchange.

Raspberry Pi-based ophthalmic imaging has precedent: Shen and Mukai demonstrated a Pi-based fundus camera [14], and the Raspberry Pi Foundation has highlighted affordable ocular imaging as an application area [13]. The **arroyo-pi** system extends this tradition to pupillography.

10 Conclusion

Arroyo’s asthenocoria sign has endured for a century as a clinical observation without instrumentation. The **arroyo-pi** system replaces the penlight-and-stopwatch exam with an embedded pupillography platform that quantifies pupillary behavior under sustained light using near-infrared illumination, OpenCV-based tracking, and a deterministic analysis pipeline.

The system is deliberately honest about its limitations. It measures pupillary behavior, not adrenal function. Its grading thresholds are functional-medicine screening bands, not validated diagnostic criteria. “Adrenal fatigue” is not a recognized endocrinological diagnosis, and the ACTH stimulation test remains the gold standard for adrenal insufficiency. Asthenocoria is nonspecific.

What the system provides is objectivity where there was subjectivity, repeatability where there was inconsistency, and a quantitative record where there was clinical impression. For practitioners who use Arroyo’s test as a screening and trend-tracking instrument, this is a meaningful improvement. The sub-\$100 hardware cost and open-source software make it accessible to the clinics and practices where the test is actually performed.

Future work should focus on clinical validation: comparing the system’s measurements against established pupillometry protocols, characterizing inter-session variability, and evaluating whether the grading bands correlate with any validated physiological markers. Until such validation exists, the system should be used as what it is—a screening instrument that quantifies an observation, not a diagnostic tool that identifies a disease.

References

- [1] Thyroid Patient Advocacy UK. Tests for adrenal fatigue you can do at home. Available at: <https://www.tpauk.com/main/article/tests-for-adrenal-fatigue-you-can-do-at-home/>

- [2] Dr. Jockers. How to test adrenal function: the iris contraction test and grading protocol. Available at: <https://drjockers.com/test-adrenal-function/>
- [3] Farlex Medical Dictionary. Asthenocoria / Arroyo sign. Available at: <https://medical-dictionary.thefreedictionary.com/asthenocoria>
- [4] IQB Spanish Medical Dictionary. Signo de ARROYO. Available at: <https://www.iqb.es/diccio/s/signo.htm>
- [5] Garma on Health. Adrenal fatigue tests you can do: the 8-inch, 45-degree protocol. Available at: <https://garmaonhealth.com/adrenal-fatigue-tests-you-can-do/>
- [6] Pulaski, S. Alpha Omega Pupil: syntonics optometry and pupillary response. Syntonics Academy. Available at: <https://csovision.org/wp-content/uploads/2020/05/Syntonics2020AOPupilShort101ZoomAlbanyPDFX.pdf>
- [7] Szabadi, E. (2018). Regulation of the pupil by the sympathetic pathways. *Journal of Neurology, Neurosurgery & Psychiatry*, PMC6305320. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6305320/>
- [8] Hall, C.A. & Chilcott, R.P. (2018). Pupillary light reflex in neurodiagnostics. PMC5872002. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5872002/>
- [9] Ulyclinic. Asthenocoria (Arroyo's sign). Available at: [https://www.ulyclinic.com/cardiovascular-disease-conditions/asthenocoria-\(arroyo's-sign\)](https://www.ulyclinic.com/cardiovascular-disease-conditions/asthenocoria-(arroyo's-sign))
- [10] AMBOSS. Adrenal insufficiency: ACTH stimulation test. Available at: <https://www.amboss.com/us/knowledge/adrenal-insufficiency>
- [11] Kuthe, S. & Lakhe, S. (2025). Smartphone pupillometry: methods and validation. PMC12391761. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12391761/>
- [12] At-home pupillometry: smartphone validation against clinical instruments. PMC10686294. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10686294/>
- [13] Raspberry Pi Foundation. An affordable ocular fundus camera. Available at: <https://www.raspberrypi.com/news/an-affordable-ocular-fundus-camera/>
- [14] Shen, A. & Mukai, S. (2017). A low-cost Raspberry Pi-based fundus camera. PMC5370517. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5370517/>
- [15] OpenIrisDPI: an open-source eye tracking and pupillometry framework. PMC12900282. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12900282/>
- [16] Dsouza, J. InfraredPupilTracker: pupil tracking on Raspberry Pi with OpenCV. GitHub repository. Available at: <https://github.com/JohnalDsouza/InfraredPupilTracker>
- [17] Arducam. IMX462 Pivariety camera documentation. Available at: <https://docs.arducam.com/Raspberry-Pi-Camera/Pivariety-Camera/IMX462/>
- [18] ams OSRAM. 940 nm IR for eye, face, and hand tracking. Available at: <https://ams-osram.com/applications/mobile-wearables/eye-face-hand-tracking>

- [19] Babcock, J.S. & Pelz, J.B. (2004). Building a lightweight eyetracking headgear. *ETRA 2004*. Available at: https://www.cis.rit.edu/people/faculty/pelz/publications/ETRA04_babcock_pelz0LD.pdf
- [20] Vecchieschi, P. et al. (2026). MEYELens: open-source pupillometry eyewear. *Behavior Research Methods*. Available at: https://ricerca.sns.it/retrieve/359895a9-ea53-4b65-8713-e75673bef581/Vecchieschi_et_al-2026-Behavior_Research_Methods.pdf
- [21] OpenIris: an open-source framework for eye tracking. ACM Digital Library. Available at: <https://dl.acm.org/doi/pdf/10.1145/3649902.3653348>
- [22] Pinn Clinic. Adrenal fatigue and Wilson’s 1998 framework. Available at: <https://www.pinnclinic.com/adrenal-fatigue>
- [23] Adrenal Fatigue Solution. Testing for adrenal fatigue: the iris contraction test. Available at: <https://adrenalfatiguesolution.com/testing-for-adrenal-fatigue/>
- [24] Clinical Gate. Naturopathic diagnostic techniques. Available at: <https://clinicalgate.com/2015/05/27/naturopathic-diagnostic-techniques/>
- [25] Darwyn Health. Naturopathic diagnostic methods: a comprehensive guide. Available at: darwynhealth.com
- [26] Kelbsch, C. et al. (2023). Pupillary escape and the Loewenfeld legacy. PMC10199107. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10199107/>
- [27] Mathôt, S. (2018). Pupillometry: a review of methods and applications. *Journal of Cognition*. Available at: <https://journalofcognition.org/articles/10.5334/joc.18>
- [28] Browning, M. et al. Pupillary measures: guidelines for researchers. *Psychophysiology*. Available at: <https://onlinelibrary.wiley.com/doi/10.1111/psyp.14035>
- [29] WebMD. Adrenal fatigue: is it real? Available at: <https://www.webmd.com/a-to-z-guides/adrenal-fatigue-is-it-real>
- [30] Clinical Education. Adrenal fatigue, adrenal insufficiency, or a misnamed state of hypofunction. Available at: <https://www.clinicaleducation.org/resources/reviews/adrenal-fatigue-adrenal-insufficiency-or-a-misnamed-state-of-hypofunction/>
- [31] Medical News Today. Adrenal fatigue myths. Available at: <https://www.medicalnewstoday.com/articles/245810>